

Byte2Cloud

Linux to Cloud Bootcamp

PRE-REQUIREMENTS GUIDE

Session 01: Getting Started with Linux

DATE	TIME
09 June 2026	06:30 PM



Organized by
Electronics and Computer Science Club (ECSC)
University of Kelaniya

Please complete pre-requirements before the session.

// Introduction

Welcome to Byte2Cloud — the Linux to Cloud Bootcamp organized by the Electronics and Computer Science Club (ECSC) at the University of Kelaniya. This bootcamp is designed to take you from zero knowledge of Linux all the way to building and managing cloud infrastructure.

To make the most of Session 01: Getting Started with Linux, you need to have a working Linux environment set up on your computer before the session begins. This document will walk you through three different methods to install Ubuntu Linux, so you can choose the one that best suits your hardware and preferences.

Important: Please complete your Linux installation before attending the session on 09 June 2026 at 06:30 PM. If you run into any issues, reach out to us via the WhatsApp group.

// System Requirements

Before proceeding, ensure your computer meets the minimum requirements:

Component	Minimum Requirement
Processor	64-bit dual-core CPU
RAM	4 GB (8 GB recommended)
Storage	20 GB free disk space
Operating System	Windows 10/11 or macOS 10.15+
Internet	Stable internet connection
Virtualization	Enabled in BIOS/UEFI (for VM/WSL)

// Download Ubuntu ISO

All three installation methods require the Ubuntu ISO image. Download it first before proceeding with any method.

Download Link: <https://ubuntu.com/download/desktop> — Ubuntu 22.04 LTS (recommended)

Download the file ending in .iso (approximately 4–6 GB). Save it to a location you can easily find, such as your Downloads folder.

// Method 1: Windows Subsystem for Linux (WSL)

What is WSL?

WSL (Windows Subsystem for Linux) is a feature built into Windows 10 and 11 that lets you run a Linux terminal directly inside Windows — no virtual machine or reboot required. It is the quickest and easiest way to get started with Linux.

Best for: Windows 10/11 users who want a fast, lightweight setup. WSL provides a full Linux command-line environment.

Step-by-Step Installation

1 Enable WSL Feature

Open PowerShell or Windows Terminal as Administrator:

- Press the Windows key, type PowerShell
- Right-click on Windows PowerShell and select Run as administrator
- Click Yes when prompted by User Account Control

```
# Enable WSL and install the default Linux distribution
wsl --install

# If you want specifically Ubuntu:
wsl --install -d Ubuntu
```

Note: This command enables the Virtual Machine Platform, Windows Subsystem for Linux, and installs Ubuntu automatically.

2 Restart Your Computer

After the installation completes, you will be prompted to restart your computer. Save all your work and restart.

3 Set Up Your Linux User Account

After restarting, Ubuntu will automatically open. Set up your Linux user account:

1. **Enter a username:** Choose a Linux username (it can be different from your Windows username, and should be lowercase)
2. **Enter a password:** Create a password (note: when typing a password in Linux, nothing is shown on screen — this is normal)
3. **Confirm password:** Re-enter your password

Once set up, verify your Ubuntu installation by running:

```
# Check Ubuntu version
lsb_release -a

# Check Linux kernel version
uname -r

# Update package lists
sudo apt update
```

Success! If you see Ubuntu version information printed, your WSL installation is working correctly.

Accessing WSL After Installation

- From the Start Menu, search for Ubuntu and click on it
- Or open Windows Terminal and select Ubuntu from the dropdown
- Or type wsl in the Run dialog (Win + R)

Tip: You can access your Windows files from WSL at the path /mnt/c/ (for your C: drive). Your Linux home folder is accessible from Windows Explorer by navigating to \\wsl\$\\Ubuntu\\home\\<username>.

// Method 2: Virtual Machine (VirtualBox / VMware)

What is a Virtual Machine?

A Virtual Machine (VM) runs a complete operating system inside a window on your existing OS. This approach gives you a full Ubuntu desktop experience without modifying your existing system. You can run Ubuntu and Windows side-by-side simultaneously.

Best for: Users who want a full Ubuntu desktop GUI, or those on Windows or macOS. Requires more RAM (8 GB+ recommended).

Option A: Using VirtualBox (Free & Open Source)

1 Download & Install VirtualBox

Download VirtualBox from the official website:

- Visit: <https://www.virtualbox.org/wiki/Downloads>
- Download the installer for your operating system (Windows / macOS / Linux)
- Run the installer and follow the on-screen instructions
- Also download and install the VirtualBox Extension Pack from the same page

2 Create a New Virtual Machine

4. **Open VirtualBox** and click New
5. **Name:** Type Ubuntu (VirtualBox will auto-detect the OS type as Linux / Ubuntu 64-bit)
6. **ISO Image:** Click the dropdown and browse to your downloaded Ubuntu .iso file
7. **Check** the box that says Skip Unattended Installation (recommended for first-time users)
8. **Click Next**

3 Configure Hardware

On the Hardware screen:

- Base Memory (RAM): Set to at least 4096 MB (4 GB). Recommended: 4096–8192 MB
- Processors: Set to at least 2 CPUs if your processor has 4+ cores
- Click Next

On the Virtual Hard Disk screen:

- Select Create a Virtual Hard Disk Now
- Set size to at least 25 GB (recommended: 30–50 GB)
- Click Next, then Finish

4 Install Ubuntu

9. **Select your VM** from the list and click Start
10. **Ubuntu will boot** from the ISO. Select Try or Install Ubuntu

11. **On the installer screen:** click Install Ubuntu
12. **Keyboard Layout:** Select your preferred layout and click Continue
13. **Installation type:** Select Normal installation, check both download boxes, click Continue
14. **When asked about disk:** Select Erase disk and install Ubuntu (this only affects the virtual disk, not your real files)
15. **Set timezone:** Select your timezone on the map
16. **Create user:** Enter your name, username, and a strong password
17. **Wait** for installation to complete (~10–20 minutes), then click Restart Now

5 Install Guest Additions (Recommended)

Guest Additions improve the VM experience (resizable screen, clipboard sharing, etc.):

```
# Inside your Ubuntu VM, open a terminal and run:
sudo apt update
sudo apt install -y virtualbox-guest-additions-iso
sudo apt install -y virtualbox-guest-utils

# Then restart your VM:
sudo reboot
```

Option B: Using VMware Workstation Player (Free for Personal Use)

- Download VMware Workstation Player from: <https://www.vmware.com/products/workstation-player.html>
- Install VMware Workstation Player and launch it
- Click Create a New Virtual Machine
- Select Use ISO image and browse to your Ubuntu .iso file
- Fill in your name, username, and password for Easy Install
- Set disk size to at least 25 GB
- Click Finish — VMware will automatically install Ubuntu

Tip: VMware Easy Install mode automates the entire Ubuntu installation. You just need to provide your credentials and it handles everything else.

// Method 3: Dual Boot (Ubuntu alongside Windows)

What is Dual Boot?

Dual booting installs Ubuntu directly on your physical hard drive alongside Windows. When you start your computer, you choose which operating system to boot into. This gives you native Linux performance but requires more care during setup.

Warning: Dual booting involves partitioning your hard drive. Incorrect setup can cause data loss. Back up all important files before proceeding. If you are unsure, choose WSL or VirtualBox instead.

Step-by-Step Installation

1 Back Up Your Data

Before anything else, create a backup of all important files on an external drive or cloud storage. This is a critical step.

2 Create a Bootable USB Drive

You will need a USB drive (8 GB or larger). All data on the USB will be erased.

Using Rufus (Windows — Recommended)

- Download Rufus from: <https://rufus.ie>
- Plug in your USB drive
- Open Rufus and select your USB drive under Device
- Click SELECT and choose your Ubuntu .iso file
- Leave all other settings as default
- Click START and confirm any warnings
- Wait for the process to complete, then click CLOSE

Using balenaEtcher (Windows / macOS / Linux)

- Download balenaEtcher from: <https://www.balena.io/etcher>
- Open Etcher and click Flash from file — select your Ubuntu .iso
- Click Select target and choose your USB drive
- Click Flash! and wait for completion

3 Create a Partition for Ubuntu

You need to free up at least 25 GB of disk space for Ubuntu. Do this from within Windows:

- 18. Right-click on This PC** or Computer and select Manage
- 19. Go to Disk Management** (under Storage in the left panel)

20. **Right-click on your main drive (usually C:)** and select Shrink Volume
21. **In the 'Enter the amount of space to shrink' box**, enter at least 25600 (for 25 GB). Enter 51200 for 50 GB (recommended)
22. **Click Shrink**. A new unallocated space will appear — leave it as-is (do not format it)

4 Boot from USB

Restart your computer with the USB drive plugged in and boot from it:

- Restart your computer
- As the computer starts, press the boot menu key. Common keys: F12 (Dell, Lenovo), F9 (HP), Option key (Mac), or Esc (some laptops)
- Select your USB drive from the boot menu
- Select Try or Install Ubuntu from the GRUB menu

Tip: If you can't access the boot menu, you may need to disable Secure Boot or enable Legacy Boot in your BIOS/UEFI settings. Access BIOS by pressing F2, Del, or F10 during startup.

5 Install Ubuntu (Dual Boot Setup)

23. **Select language** and click Install Ubuntu
24. **Choose keyboard layout** and click Continue
25. **Select Normal Installation** and check both option boxes, click Continue
26. **IMPORTANT — Installation type:** Select "Install Ubuntu alongside Windows Boot Manager". This is the safe dual-boot option. Do NOT select "Erase disk" as this will delete Windows
27. **Drag the divider** to allocate disk space between Windows and Ubuntu (give Ubuntu at least 25 GB)
28. **Click Install Now** and confirm the changes
29. **Select your timezone** on the map
30. **Create your user account:** enter your name, username, and a strong password
31. **Wait** for installation to complete (~15–30 minutes)
32. **Click Restart Now** when prompted and remove the USB drive

6 Boot Into Ubuntu

After restarting, you will see the GRUB bootloader menu — this appears every time you start your computer. Use the arrow keys to select Ubuntu or Windows Boot Manager and press Enter.

Success! If you see the Ubuntu login screen, your dual boot installation is complete. Enter your password to log in.

// Method Comparison

Not sure which method to choose? Use this table to decide:

Method	Works On	Difficulty	Overhead	Experience	Speed
WSL	Windows only	Very Easy	Low	Command-line only	Fastest
VirtualBox	Any OS	Easy	Medium	Full GUI desktop	Moderate
VMware	Any OS	Very Easy	Medium	Full GUI desktop	Moderate
Dual Boot	Any OS	Advanced	None (native)	Full GUI desktop	Native

Our Recommendation: For most participants, WSL (Method 1) is the fastest to set up and perfectly suited for the bootcamp content. If you want the full Linux desktop experience, go with VirtualBox (Method 2).

// Post-Installation Checklist

After installing Ubuntu using any method, run these commands to make sure your system is ready for the session:

```
# 1. Update package lists and upgrade installed packages
sudo apt update && sudo apt upgrade -y

# 2. Install essential tools
sudo apt install -y curl wget git vim nano net-tools

# 3. Check your internet connection
ping -c 4 google.com

# 4. Verify your Ubuntu version
lsb_release -a

# 5. Check available disk space
df -h
```

// Common Issues & Troubleshooting

Issue	Solution
WSL: 'wsl --install' not recognized	Your Windows version may be older. Run: wsl --install in PowerShell as Admin, or update Windows first via Settings > Windows Update.
VirtualBox: Virtualization disabled	Enter BIOS/UEFI and enable Intel VT-x or AMD-V. The key to enter BIOS is usually F2, Del, or F10 during startup.
WSL: No internet inside Linux	Run: wsl --shutdown then restart WSL. Also check if your antivirus or firewall is blocking WSL.
Ubuntu installer freezes	Try a different USB port or recreate the bootable USB. Verify the ISO file was downloaded completely.
GRUB not appearing after dual boot	Boot from your USB again and use the Boot Repair tool. Or contact us on WhatsApp for help.
Black screen in VirtualBox	Enable 3D Acceleration in VM Settings > Display, and increase Video Memory to 128 MB.

// Need Help? Contact Us

If you face any issues setting up your environment, we are here to help. Reach out before the session so we can assist you in time.

WhatsApp: Join the Byte2Cloud WhatsApp group for real-time updates and support. The link will be shared via the ECSC social media channels listed above.

We look forward to seeing you at Byte2Cloud Session 01 on 09 June 2026!
Electronics and Computer Science Club (ECSC) | University of Kelaniya